

$$\Rightarrow B=A \text{ (Theorem 2(e))}.$$

Let, there exist an $n \times n$ identity matrix I_n such that $CA^{-1} = I_n$.

$$\Rightarrow (CA^{-1}) \cdot A = I_n \cdot A \Rightarrow C \cdot (A^{-1}A) = I_n \cdot A \text{ (Theorem 2(a))}$$

$$\Rightarrow C \cdot I_n = I_n \cdot A \text{ (By defn. of inverse)} \Rightarrow C = A \text{ (Theorem 2(e))}.$$

$\therefore A$ is the unique matrix s.t. $A^{-1}A = AA^{-1} = I_n$.

$$\text{By defn. of inverse, } \boxed{(A^{-1})^{-1} = A} \quad \square$$

b) By definition of inverse, there exists a unique matrix A^{-1} and B^{-1} s.t.
 $A^{-1}A = AA^{-1} = I_n \quad B^{-1}B = BB^{-1} = I_n$.

$$AA^{-1} = I_n \Rightarrow (AI_n)A^{-1} = I_n \text{ (Theorem 2(e))}$$

$$\Rightarrow (A(BB^{-1}))A^{-1} = I_n \text{ (Given)} \Rightarrow (AB)B^{-1}A^{-1} = I_n \text{ (Theorem 2(a))}$$

$$\Rightarrow (AB)(B^{-1}A^{-1}) = I_n \text{ (Theorem 2(a))}$$

$$B^{-1}B = I_n \text{ (Given)} \Rightarrow (B^{-1}I_n)B^{-1} = I_n \text{ (Theorem 2(e))}$$

$$\Rightarrow (B^{-1}(A^{-1}A))AB = I_n \text{ (Given)} \Rightarrow ((B^{-1}A^{-1})A)B = I_n \text{ (Theorem 2(a))}$$

$$\Rightarrow (B^{-1}A^{-1})(AB) = I_n \text{ (Theorem 2(a))}$$

$\therefore$ There exists a matrix $B^{-1}A^{-1}$ s.t. $\boxed{(AB)(B^{-1}A^{-1}) = (B^{-1}A^{-1})(AB) = I_n}$
 (Existence part proved).

Let, there exists a $n \times n$ matrix X s.t. $(AB)X = I_n$.

$$\therefore A^{-1}((AB)X) = A^{-1} \cdot I_n \Rightarrow (A^{-1}(AB))X = A^{-1} \cdot I_n \text{ (Theorem 2(a))}$$

$$\Rightarrow ((A^{-1}A)B)X = A^{-1} \cdot I_n \text{ (Theorem 2(a))} \Rightarrow (I_n B)X = A^{-1} \cdot I_n$$

$$\text{(By defn. of inverse)} \Rightarrow BX = A^{-1} \text{ (Theorem 2(e))} \Rightarrow B^{-1}(BX) = B^{-1}A^{-1}$$

$$\Rightarrow (B^{-1}B)X = B^{-1}A^{-1} \text{ (Theorem 2(a))} \Rightarrow I_n X = B^{-1}A^{-1} \text{ (By defn. of inverse)}$$

$$\Rightarrow \boxed{X = B^{-1}A^{-1} \text{ (Theorem 2(e))}}$$

Let, there exists a $n \times n$ matrix Y s.t. $Y(AB) = I_n$.

$$\therefore (Y(AB)) \cdot B^{-1} = I_n \cdot B^{-1} \Rightarrow Y \cdot ((AB) \cdot B^{-1}) = I_n \cdot B^{-1} \text{ (Theorem 2(a))}$$

$$\Rightarrow Y \cdot (A \cdot (BB^{-1})) = I_n \cdot B^{-1} \text{ (Theorem 2(a))} \Rightarrow Y \cdot (A \cdot I_n) = I_n \cdot B^{-1}$$

(By defn. of inverse)